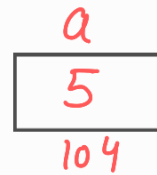
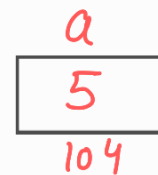
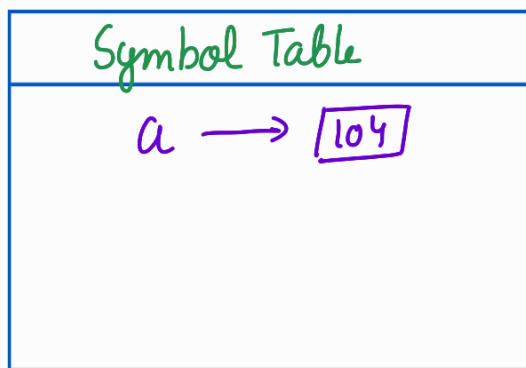


Suppose we have a variable 'a' as `int a = 5;`

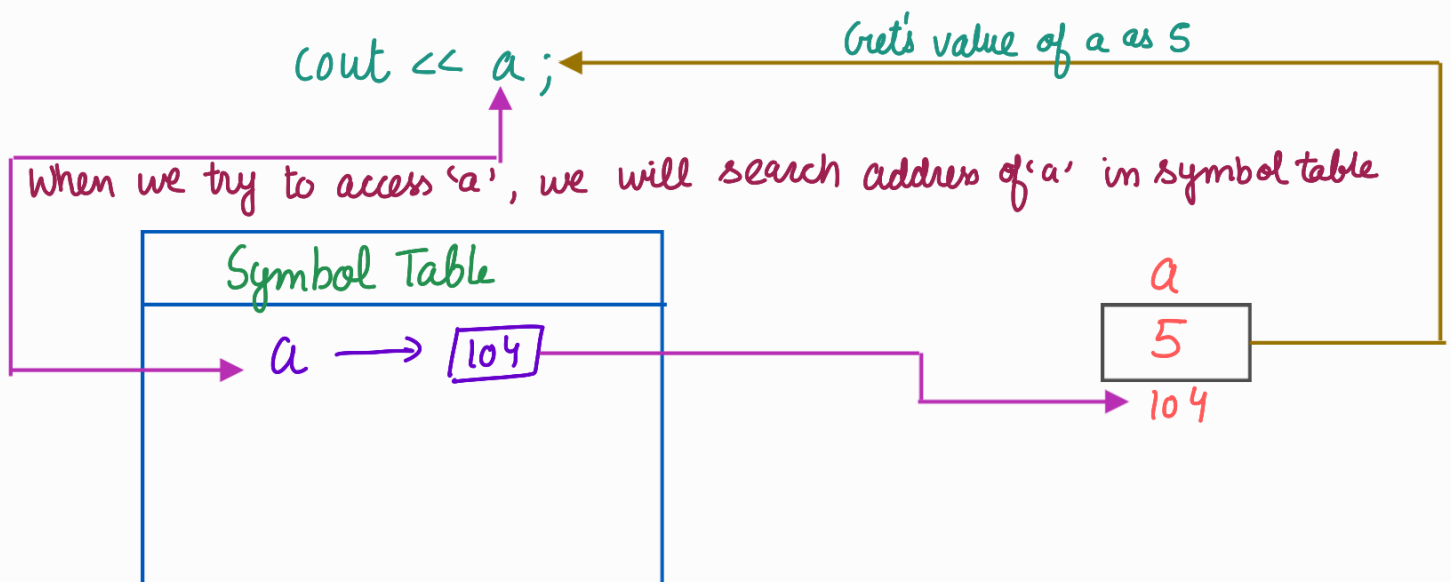
Then this '5' will be stored at a certain location in addr.



Now all these variables & their address are stored in a Symbol Table



Then if we want to access the variable then we can do that by using

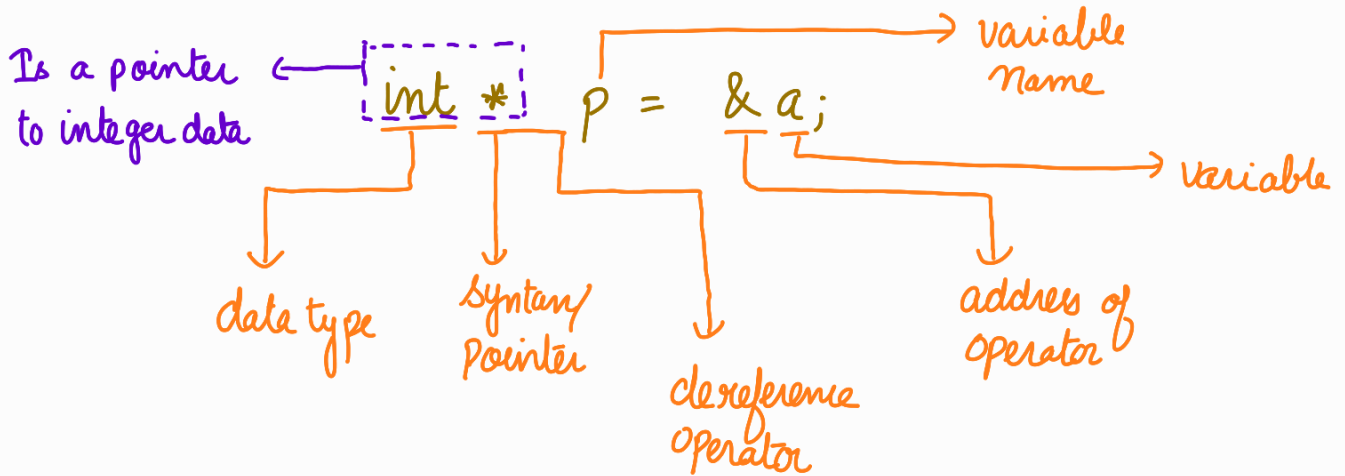


To get the address of a variable, we can use '&a'

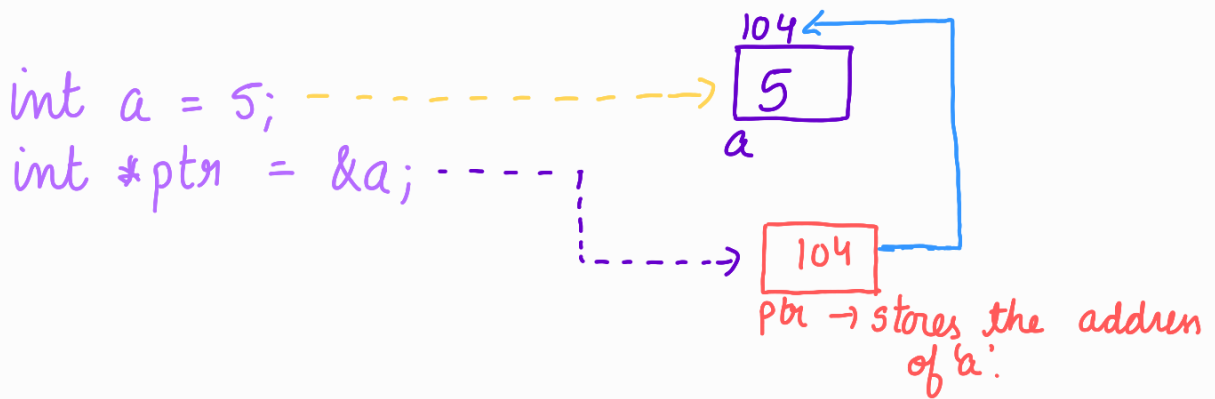
Pointers

Datatype which stores the address of the variable.

int a = 5;



Ex-2 `Char * p;`
↳ `p` is a pointer to char data



With `ptr` we can have access to 2 values:-

`ptr` → address
 → value



`cout << ptr << endl;` // gives 104 (value stored at `ptr`)
`cout << *ptr << endl;` // gives value at address stored at `ptr` i.e '5'.

To access

- ↳ get addr of 'a' → use 'p'
- ↳ value → use '*p'
- ↳ get address of p → use '&p'

- 1) *ptr = value stored at location in ptr
- 2) &ptr → address of ptr
- 3) &a → addr of a
- 4) a → value of a
- 5) ptr → value of ptr

→ int a = 5; int *p = &a → size of p → 8
char ch = 'b'; char *c = &ch; → size of p → 8
double d = 1.03; double *dtr = &d → size of p → 8

↓
Size will be same as it only stores the addr type

Why we need pointer?

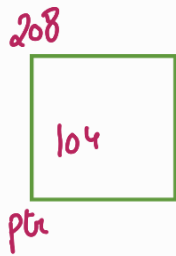
- ↳ Dynamic memory allocation
- ↳ Use in hardware access
- ↳ memory management

Note → int *ptr; // Bad practise

cout << *ptr << endl; → We will get Segmentation fault as it tries to access memory which doesn't exist.

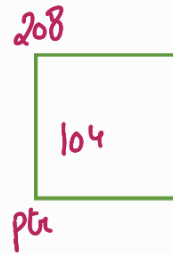
We have to create null ptr to avoid error → int *ptr = 0; or
int *ptr = NULL; or int *ptr = nullptr;

Pointer arithmetic :-



int a = 5
int *ptr = &a;

$$a = a + 1 = 5 + 1 = 6$$



$$ptr = ptr + 1 = 108$$

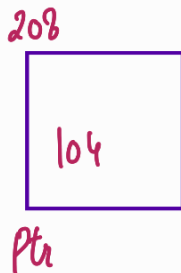
int byte = 4 bits (104 + 4 = 108)



$$*ptr = *ptr + 1 = 5 + 1 = 6$$

↓
value at address
stored in p

Ex-11



a → 10
&a = 104
ptr = 104
*ptr = 10
&ptr = 208

$$\rightarrow *ptr * 2 = 20$$

$$\rightarrow *ptr = *ptr * 2 = 30$$

$$\rightarrow (*ptr)++ = 11$$

$$\rightarrow *ptr = \frac{*ptr}{2} = 15$$

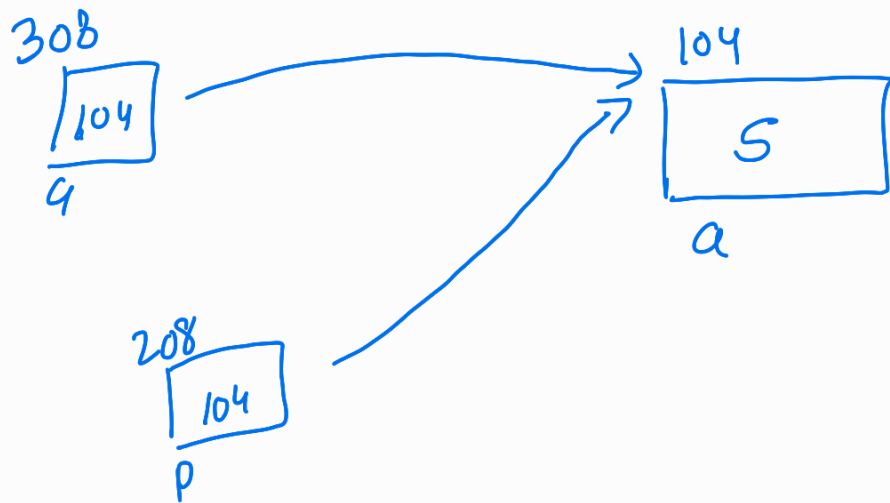
$$\rightarrow ++(*ptr) = 12$$

$$\rightarrow a = a + 1 = 13$$

$$\rightarrow *ptr = *ptr + 2 = 15$$

Copy Pointer $\rightarrow$ int *ptr2 = ptr $\rightarrow$ ptr is copied to ptr2

int a = 5
int *p = &a
int *q = p



- a → 5
- &a → address of a
- p → 104
- &p → 208
- *p → 5
- q → 104
- &q → 308
- *q → 5
- ∴ *p/2 = 5/2
- ∴ *q/2 = 5/2